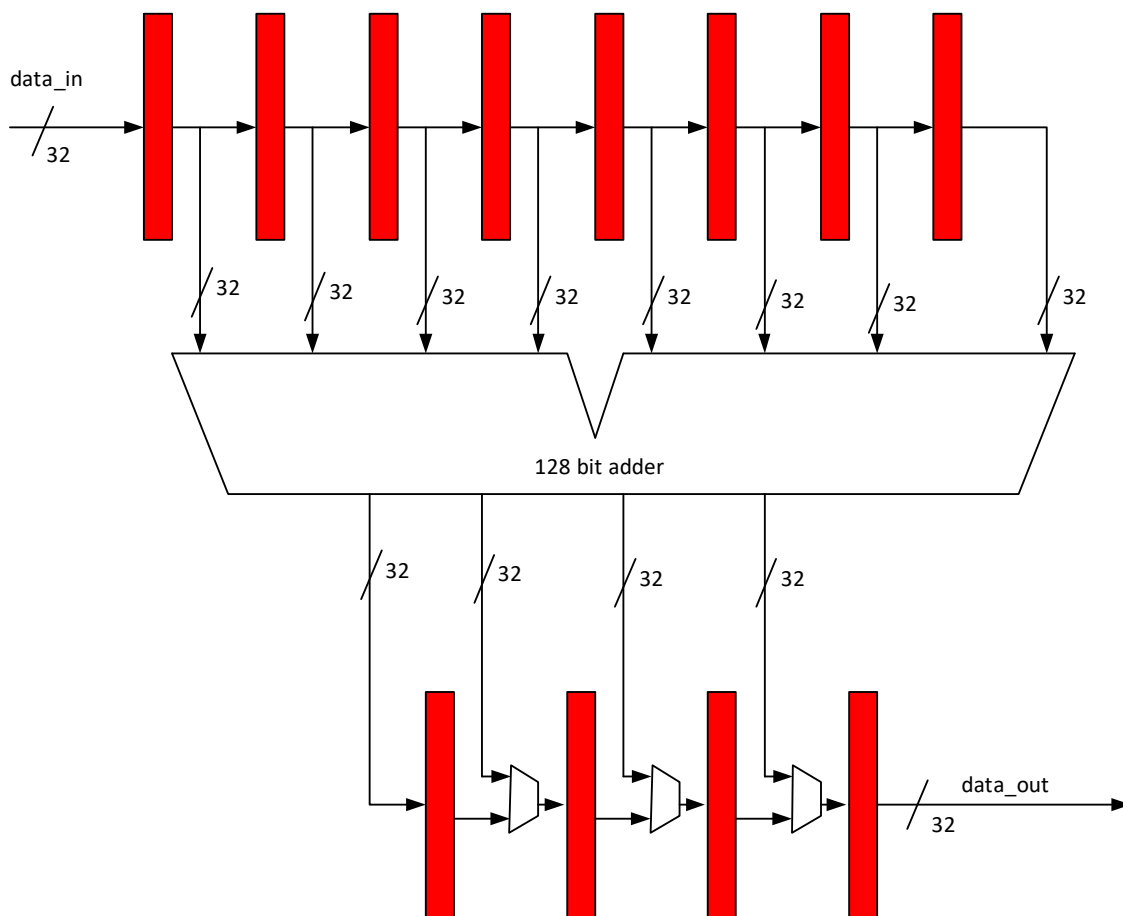


TFE4141 Design of Digital Systems 1

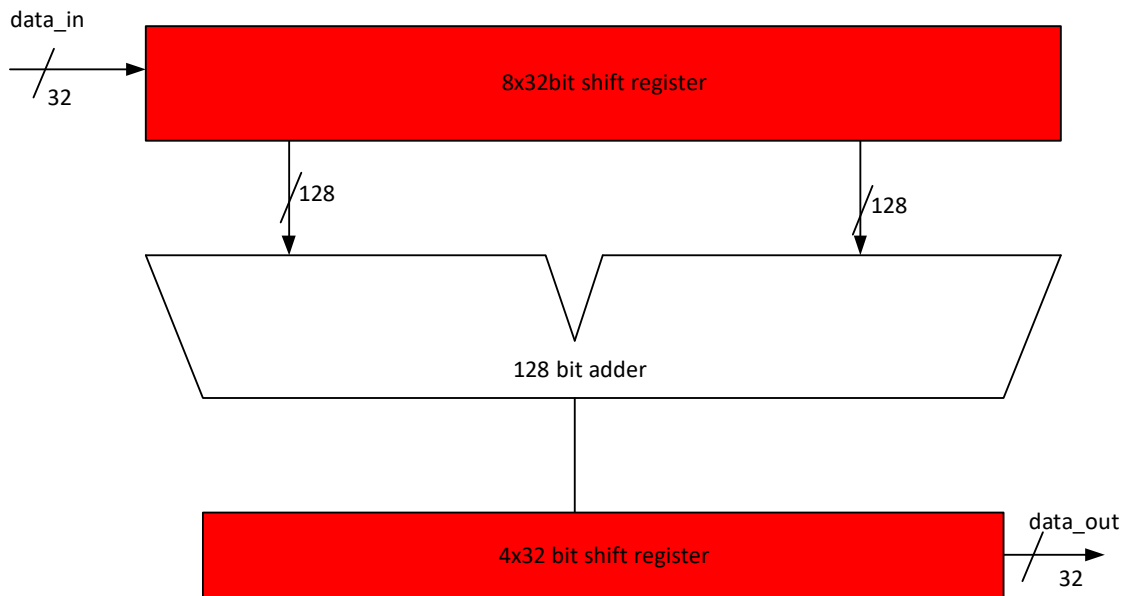
SOLUTION EXAMPLE: Assignment 4: Datapath design (Task 1 – 4)

Task 1: Draw microarchitecture

Directly from the VHDL code:



A more conceptual understanding:



The control part of the adder keeps track of the interface signals and on the number of 32 bit data chunks that have been shifted in and out as well as when the actual 128 bit addition can be performed, i.e., saved to the output register.

Task 2: Simulate the circuit

Should be straightforward with the attached testbench.

Task 3: Count the number of flip-flops inferred during synthesis

Datapath: $32 \times 12 \text{ bit} = 384 \text{ bit}$.

Control logic: 6 bit

Task 4: Synthesize the design and report the max clock frequency

Max frequency: 166 MHz.

The longest path goes through the adder.

Hint for Task 5: Since the critical path is through the very long 128 bit adder, try to find ways to reduce the adder size.